Unit 1: Variables and Expressions

A variable is a symbol, usually a letter, that represents an unknown number. An algebraic expression combines variables, numbers, and operations such as addition and multiplication. Evaluating an expression means substituting a value for each variable and then computing the result.

Unit 2: Linear Equations

A linear equation is an equation whose graph is a straight line. The slope-intercept form is written $y = mx + b$, where m is the slope and b is the y-intercept. The slope measures how steep the line is, and the y-intercept is the point where the line crosses the y-axis.

Unit 3: Systems of Equations

A system of equations is a set of two or more equations that share the same variables. A solution to the system is a point that satisfies every equation at once. Systems can be solved by substitution or by elimination.

Unit 4: Quadratic Functions

A quadratic function has the form $y = ax^2 + bx + c$, where a is not zero. Its graph is a smooth curve called a parabola. The highest or lowest point of the parabola is called the vertex.

Unit 5: Solving Quadratic Equations

The roots of a quadratic are the x-values where the parabola crosses the x-axis. The quadratic formula gives the roots of any quadratic equation. The expression $b^2 - 4ac$ is called the discriminant, and it tells how many real roots the equation has.

Unit 6: Exponents and Polynomials

An exponent tells how many times a base is multiplied by itself. A polynomial is a sum of terms, where each term is a number times a variable raised to a whole-number power. The degree of a polynomial is the largest exponent that appears in it.